Grade 11 Term 2 Home Practice – C2

Point estimates and confidence levels

Global Economics – 2026–2027

Contents

Evaluated criteria 1

1 Introduction 1

2 Notation Reference and Practice Procedure 4

3 General Worked Problems 5

4 Advanced Problems 11

Evaluated criteria

C2 Compares point and interval estimation and interprets confidence levels through their long-run coverage meaning, distinguishing this from the probability that a particular calculated interval contains the parameter.

How to practise. Try each task before opening its blue solution link. Use the arrows to return to the contents, problem list, or problem. In Task 1, calculate the standard error, then calculate and interpret the confidence intervals. In Task 2, use the evidence to choose one answer.

1 Introduction

Problems:

1.1. Key terms and highlight guide

1.2. From cycling trips to an interval

1.1 Key terms and highlight guide

A **population parameter** describes the population; here, the population mean is fixed. A **point estimate** uses a sample statistic to estimate that parameter. An **interval estimate** gives a range of plausible parameter values under the model. The **confidence level** describes the interval procedure's long-run coverage.

Use the following table as a guide to the terms, symbols, and highlights used throughout this document. Greens connect sample quantities, and reds connect population quantities. Two blue shades distinguish the confidence level (central area) from its critical value. Separate warm shades distinguish standard error, margin of error, endpoints, and full width; names and symbols identify each quantity even without color.

Terms and symbols	Meaning in context
Population mean, μ	The fixed population parameter being estimated.
Point estimate, $\bar{x}$	Sample mean and interval centre; for example, 60 units.
Population SD, σ; sample size, n	σ describes individual observations; n counts sampled units. For example, 20 units and 100 give SE $= 20/\sqrt{100} = 2$ units.
Standard error, SE	Spread of sample means across repeated samples, or its estimate; for example, 2 units. Distinct from individual-observation spread.
Margin of error, MoE	Distance from centre to an endpoint: MoE $= z^*$ SE. For example, 1.96 (2) = 3.92 units.
Endpoints, $\bar{x} \pm$ MoE	Lower and upper bounds: 60 $\pm$ 3.92 gives 56.08 and 63.92 units.
Full width, 2 MoE	Distance between endpoints, twice the margin of error; here 2(3.92) = 7.84 units.
Confidence level, C	Nominal long-run coverage under the model; 95% = 0.95 is the central normal area.
Standard normal critical value, z^*	Unitless multiplier selected by C; 1.96 bounds the central area 0.95 between $-z^*$ and z^*.
Observed coverage, rate	Fraction covering the fixed mean in a run; for example, 4/5 = 80%. Each interval covers or misses; this fraction can differ from C.



1.2 From cycling trips to an interval

Is the average weekly cycling trip to the market under 42 minutes? A market planner wants to find out. A survey of cyclists gave a sample mean of 40 minutes. This *point estimate* is below 42 minutes, but how much confidence should the planner place in that comparison?

The survey's 95% confidence interval runs from 36 to 44 minutes. This range gives plausible values for the population mean, the average trip time across the whole population. Individual trip times can vary and can fall outside this range. Since the interval includes 42 minutes and values above it, the planner cannot conclude from this interval that the population mean is under 42 minutes.

A fresh sample could give a different point estimate and different interval endpoints, while the population mean stays fixed. If we repeated the same procedure with many samples under the model, about 95 out of every 100 intervals would contain that fixed mean. What makes an interval wider or narrower, and how should we interpret its confidence level?



2 Notation Reference and Practice Procedure

For the normal procedures used here, let n be the sample size and σ be the known population standard deviation. MoE stands for *margin of error*, the distance from the sample mean to either endpoint.

Notation	Meaning
μ	Population mean, the fixed parameter of interest
$\bar{x}$	Sample mean, a point estimate of μ
$\mathrm{SE} = \sigma/\sqrt{n}$	Standard error of $\bar{x}$ for independent observations
z^*	Standard normal critical value for the confidence level
$\mathrm{MoE} = z^* \mathrm{SE}$	Margin of error at the stated confidence level
$[\bar{x} - \mathrm{MoE}, \bar{x} + \mathrm{MoE}]$	Two-sided confidence interval, with width $2\,\mathrm{MoE}$

Use the same units for μ, $\bar{x}$, SE, MoE, the endpoints, and the width.

2.1 Choosing the critical value

Write the confidence level C as a decimal. On the standard normal curve, the central area from $-z^*$ to z^* is C, leaving $(1-C)/2$ in each tail. The area to the left of z^* is then $C + (1-C)/2 = (1+C)/2$, so

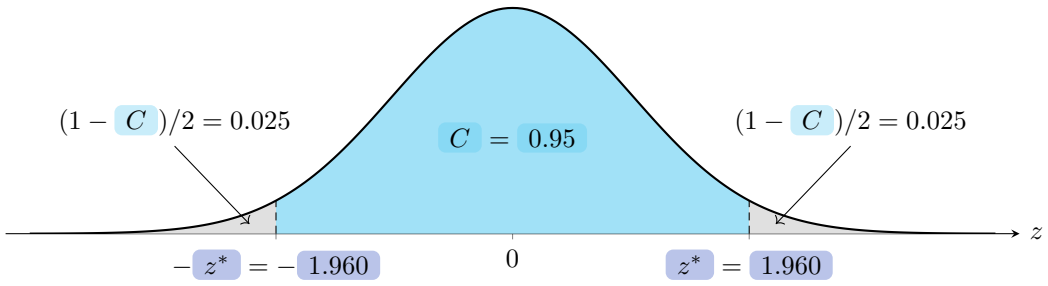
$$z^* = \Phi^{-1}\left(\frac{1+C}{2}\right).$$

Φ gives the area to the left of a value on the standard normal curve. Its inverse, Φ^{-1}, takes an area and finds the value with that much area to its left. For 95% confidence, each tail has area 0.025, so look for 0.975 to the left of z^*.

Choose z^* from the confidence level and calculate $\bar{x}$ from the sample data. Higher confidence requires a larger z^* to enclose more area. With the same standard error, this gives a larger MoE and a wider interval.

Confidence level	Standard normal critical value z^*
80%	1.282
90%	1.645
95%	1.960
97.5%	2.241
99%	2.576
99.9%	3.291

Two-sided values rounded to three decimals.



The cumulative area to the left of the axis position $z^* = 1.960$ is the left tail plus the central region, so $\Phi(1.960) \approx 0.025 + 0.95 = 0.975$.

2.2 Practice procedure

1. **Identify the estimates.** Name the population, the mean of interest, and its units. Use $\bar{x}$ as a single estimate of μ and the interval as a range of plausible values for μ under the model.
2. **Calculate and compare intervals.** Compute SE $= \sigma/\sqrt{n}$ from the known population standard deviation and sample size. Multiply by z^* to find the margin of error, then use $\bar{x} \pm$ MoE for the endpoints. Before rounding, the width is 2 MoE. For a rounded interval, subtract the rounded lower endpoint from the rounded upper endpoint.
3. **Measure observed coverage.** In a simulation with known μ, count the intervals whose endpoints enclose μ, including a match at either endpoint. Divide that count by the total.
4. **Interpret confidence.** The nominal level is the level chosen for the procedure. It gives the approximate long-run coverage under the model. Observed coverage can vary across runs.

3 General Worked Problems

Problems:

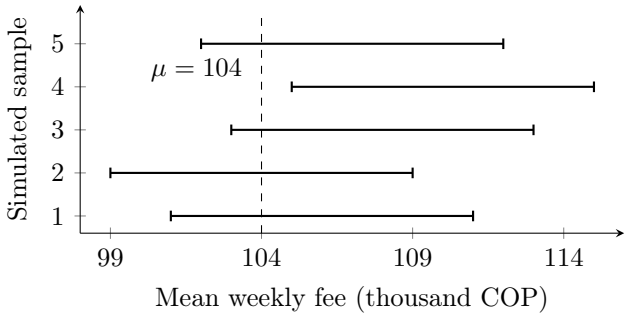
3.1. Repeated estimates of market fees

3.2. Two reports on household energy bills

3.3. Bus commute times in a simulation

3.1 Repeated estimates of market fees

A city takes a random sample of $n = 100$ market stalls to estimate the mean weekly stall fee. In thousand COP, the sample mean is $\bar{x} = 106$ and the known population standard deviation is $\sigma = 1250/49$. Assume independent observations and normally distributed sample means. The 95% confidence level uses the standard normal critical value $z^* = 1.960$.
The diagram shows the 95% intervals from five fresh samples, each of size $n = 100$, from a simulated market with mean $\mu = 104$.



Questions/Tasks

C2 Task 1. Identify the point estimate for each study sample and the population parameter being estimated. Calculate the standard error for each sample, then find the margin of error and both endpoints for each interval. Interpret the point and interval estimates in context.

C2 Task 2. Using the diagram, which option gives the correct number of intervals that cover the simulated mean and the correct interpretation of the city's 95% confidence level?

A. Four of five cover, so the observed 80% is the city's long-run coverage rate.

B. Four of five cover. The procedure covers the mean about 95% of the time in the long run under the model. The city's calculated interval either covers its fixed mean or misses it.

C. Four of five cover. The city's calculated interval has a 95% probability of covering its fixed mean.

D. Five of five cover. The procedure covers the mean about 95% of the time in the long run under the model. The city's calculated interval either covers its fixed mean or misses it.

Task 1 solution. Estimate the mean market fee

Quantity	Calculation and result	Meaning in context
Population parameter	μ, the mean weekly stall fee for all market stalls in the city, in thousand COP.	The fixed, unknown mean the city wants to estimate.
Point estimate	Sample mean $\bar{x} = 106$ thousand COP.	A single estimate of the city's mean weekly stall fee.
Standard error	SE $= \sigma/\sqrt{n} = (1250/49)/\sqrt{100} = 125/49$ thousand COP.	The standard deviation of the sample mean across repeated samples.
95% interval	MoE $= z^* \text{ SE} = 1.960\,(125/49) = 5$ thousand COP. The interval is $[106 - 5, 106 + 5] = [101, 111]$ thousand COP.	A range of plausible values for the city's mean weekly stall fee under the model.



Task 2 solution. Read the coverage diagram

Correct option B. Each simulated interval is 10 thousand COP wide. Samples 1, 2, 3, and 5 cross the line at 104. Sample 4 starts at 105 and misses, so the observed coverage is $4/5 = 80\%$. The nominal level is 95%. Over many fresh samples under the model assumptions, about 95% of the resulting intervals would contain the fixed mean.

The claim of a 95% probability for the city's interval fails. Its endpoints, 101 and 111, are fixed after sampling. The city's unknown mean either lies between them or falls outside them.



3.2 Two reports on household energy bills

A utility takes a random sample of $n = 100$ household accounts from one billing month. Both reports use this sample, with mean $\bar{x} = 120$ thousand COP. The known population standard deviation is $\sigma = 38$ thousand COP. Assume independent observations and a normal sampling model. The table gives the confidence level and standard normal critical value for each report.

Nominal confidence level	Critical value z^*	Interval width (thousand COP)
90%	1.645	
99%	2.576	

The reports round endpoints to the nearest thousand COP. The confidence levels refer to the procedures before rounding.

Questions/Tasks



C2 Task 1. Identify the point estimate for each study sample and the population parameter being estimated. Calculate the standard error for each sample, then find the margin of error and both endpoints for each interval. Interpret the point and interval estimates in context. Use the rounded endpoints to fill in the interval widths in the table.

C2 Task 2. Which option correctly compares the widths in the table and interprets the 99% confidence level?

A. The 99% interval is wider, and it contains 99% of individual bills.

B. The 99% interval is narrower, and the 99% procedure covers the population mean about 99% of the time in the long run under the model. The confidence level gives no percentage of individual bills within the reported endpoints.

C. The 99% interval is wider, and its reported endpoints have a 99% probability of enclosing the population mean.

D. The 99% interval is wider, and the 99% procedure covers the population mean about 99% of the time in the long run under the model. The confidence level gives no percentage of individual bills within the reported endpoints.

Task 1 solution. Calculate and interpret both intervals

Quantity	Calculation and result	Meaning in context
Population parameter	μ, the mean energy bill for all the utility's household accounts in that billing month, in thousand COP.	The fixed, unknown mean estimated in both reports.
Point estimate	Sample mean $\bar{x} = 120$ thousand COP.	A single estimate of the mean household bill for the month.
Standard error	SE $= \sigma/\sqrt{n} = 38/\sqrt{100} = 3.8$ thousand COP.	Both reports use the same sample and standard error.
90% interval	MoE $= z^* \text{ SE} = 1.645(3.8) = 6.251$ thousand COP. The endpoints from 120 ± 6.251 are 113.749 and 126.251. The rounded interval is [114, 126], with width $126 - 114 = 12$ thousand COP.	The narrower reported range of plausible values for the mean household bill under the model.
99% interval	MoE $= z^* \text{ SE} = 2.576(3.8) = 9.7888$ thousand COP. The endpoints from 120 ± 9.7888 are 110.2112 and 129.7888. The rounded interval is [110, 130], with width $130 - 110 = 20$ thousand COP.	The wider reported range of plausible values for the same mean under the model.

Task 2 solution. Compare confidence and width

Correct option D. Before rounding, the widths are $2(6.251) = 12.502$ and $2(9.7888) = 19.5776$ thousand COP. The reported widths are $126 - 114 = 12$ and $130 - 110 = 20$ thousand COP. With the same standard error, the larger critical value for 99% gives a larger MoE and puts the endpoints farther from 120.

Under the model assumptions, about 90% and 99% of the respective unrounded intervals would cover the fixed mean over repeated samples. Each calculated interval either covers the mean or misses it. The claim about 99% of individual bills fails because coverage concerns the population mean. The proportion of bills from 110 to 130 thousand COP depends on the distribution of individual bills.

3.3 Bus commute times in a simulation

A transit agency takes an independent random sample of $n = 36$ weekday passenger trips on one bus route. The sample mean is $\bar{x} = 34$ minutes. Assume normally distributed trip times with known population standard deviation $\sigma = 12$ minutes. The 90% confidence level uses the standard normal critical value $z^* = 1.645$.

To study the 90% procedure, the agency simulates 20 fresh samples, each of size $n = 36$, from a normal model with mean 35 minutes and the same standard deviation. The table groups the simulated intervals by sample number.

Sample numbers	1–5	6–10	11–15	16–20
Intervals containing 35 minutes	5	4	3	5

Questions/Tasks



C2 Task 1. Identify the point estimate for each study sample and the population parameter being estimated. Calculate the standard error for each sample, then find the margin of error and both endpoints for each interval. Interpret the point and interval estimates in context.

C2 Task 2. Which option correctly totals the covering intervals in the table and assesses the claim that every run of 20 must have exactly 18 intervals that cover the mean?

A. Seventeen of 20 cover. The claim fails because the 90% procedure can give different counts in different runs of 20.

B. Eighteen of 20 cover. The claim fails because the 90% procedure can give different counts in different runs of 20.

C. Seventeen of 20 cover. The claim fails because the observed 85% is the procedure's long-run coverage rate.

D. Seventeen of 20 cover. The claim fails because each calculated interval has a 90% probability of containing the fixed mean.

Task 1 solution. Estimate the mean trip time

Quantity	Calculation and result	Meaning in context
Population parameter	μ, the mean duration of all weekday passenger trips on the route, in minutes.	The fixed, unknown mean the transit agency wants to estimate.
Point estimate	Sample mean $\bar{x} = 34$ minutes.	A single estimate of the mean weekday trip time on the route.
Standard error	SE $= \sigma/\sqrt{n} = 12/\sqrt{36} = 2$ minutes.	The standard deviation of the sample mean across repeated samples.
90% interval	MoE $= z^* \text{ SE} = 1.645(2) = 3.29$ minutes. The endpoints from 34 ± 3.29 give $[30.71, 37.29]$ minutes.	A range of plausible values for the route's mean weekday trip time under the model.

Task 2 solution. Interpret a short run

Correct option A. The table gives $5+4+3+5 = 17$ intervals that contain 35 minutes, for observed coverage of $17/20 =$ 85%. This run disproves the claim of exactly 18 covering intervals in every run.

The nominal level is 90%. Under the model assumptions, about 90% of intervals cover the fixed mean over many repetitions. This gives an average of about $0.90(20) = 18$ covering intervals per run of 20, with counts that can vary from run to run. Each calculated interval either contains the mean or misses it.

4 Advanced Problems

Problems:

4.1. Paired intervals for delivery times
4.2. Sample size and produce spending

4.1 Paired intervals for delivery times

A cooperative takes an independent random sample of $n = 36$ orders delivered in October. Their sample mean is $\bar{x} = 48$ minutes. Assume normal delivery times with known population standard deviation $\sigma = 12$ minutes. Both intervals use the same sample. The standard normal critical value is $z^* = 1.645$ for 90% confidence and $z^* = 1.960$ for 95% confidence.

In a simulation with a known mean and the same standard deviation, the cooperative computes a 90% and a 95% interval from each of 100 fresh samples of size $n = 36$. The row and column labels give the nominal levels. The cells count the observed outcomes for the paired intervals.

	95% interval covers	95% interval misses
90% interval covers	89	0
90% interval misses	7	4

Questions/Tasks



C2 Task 1. Identify the point estimate for each study sample and the population parameter being estimated. Calculate the standard error for each sample, then find the margin of error and both endpoints for each interval. Interpret the point and interval estimates in context.

C2 Task 2. Which option gives the observed coverage rates for the 90% and 95% procedures, in that order, and correctly uses the way each pair is constructed to judge whether a 90% interval could cover the mean while its 95% partner misses?

A. 90% and 95%. The 95% partner must cover whenever the 90% interval covers because it is wider with the same centre.

B. 89% and 96%. The 95% partner can miss even when the 90% interval covers because some 95% intervals miss the mean.

C. 89% and 96%. The 95% partner must cover whenever the 90% interval covers because it is wider with the same centre.

D. 89% and 89%. The 95% partner must cover whenever the 90% interval covers because it is wider with the same centre.

Task 1 solution. Calculate and interpret the two ranges

Quantity	Calculation and result	Meaning in context
Population parameter	μ, the mean delivery time of all the cooperative's orders delivered in October, in minutes.	The fixed, unknown mean estimated by both intervals.
Point estimate	Sample mean $\bar{x} = 48$ minutes.	A single estimate of the mean October delivery time.
Standard error	SE $= \sigma/\sqrt{n} = 12/\sqrt{36} = 2$ minutes.	Both intervals use the same sample and standard error.
90% interval	MoE $= z^* \text{ SE} = 1.645(2) = 3.29$ minutes. The endpoints from 48 ± 3.29 give $[44.71, 51.29]$ minutes.	The narrower range of plausible values for the mean October delivery time under the model.
95% interval	MoE $= z^* \text{ SE} = 1.960(2) = 3.92$ minutes. The endpoints from 48 ± 3.92 give $[44.08, 51.92]$ minutes.	A wider range of plausible values for the same mean under the model, extending farther on both sides of 48.

Task 2 solution. Explain the coverage pattern

Correct option C. The observed coverage for the 90% procedure is $(89 + 0)/100 = 89\%$ from the covering row. For the 95% procedure it is $(89 + 7)/100 = 96\%$ from the covering column.

Within each pair, the intervals share a centre and standard error. The larger critical value for 95% gives a larger MoE, so the 95% interval contains the entire 90% interval. The 95% partner therefore covers whenever the 90% interval covers, which accounts for the zero in the table.

Under the model assumptions, about 90% and 95% of the respective intervals would cover the fixed mean over repeated sampling. Each calculated interval either contains the mean or misses it.

4.2 Sample size and produce spending

A municipal survey estimates the mean amount households in one district spent on fresh produce during a specified week. Two teams take separate random samples. Assume independent household observations and normally distributed spending with known population standard deviation $\sigma = 50$ thousand COP. Both teams use the standard normal critical value $z^* = 1.960$ for 95% confidence.

Team	Sample size n	Mean $\bar{x}$ (thousand COP)
A	25	205
B	100	205

The teams simulate 200 fresh samples at each size from a population with a known mean and the same standard deviation. Team A records 192 intervals that cover the mean. Team B records 187.

Questions/Tasks



C2 Task 1. Identify the point estimate for each study sample and the population parameter being estimated. Calculate the standard error for each sample, then find the margin of error and both endpoints for each interval. Interpret the point and interval estimates in context. Use your calculated intervals to compare widths in Task 2.

C2 Task 2. Which option correctly compares Team B's interval width with Team A's, gives the observed coverage rates for A and B, and assesses the claim that Team A's higher observed rate shows better long-run coverage?

A. B's interval is half as wide. A covers 96% and B covers 93.5% in the simulations, so A has better long-run coverage.

B. B's interval is half as wide. A covers 96% and B covers 93.5% in the simulations. Both procedures cover about 95% of the time in the long run under the model.

C. B's interval is one-quarter as wide. A covers 96% and B covers 93.5% in the simulations. Both procedures cover about 95% of the time in the long run under the model.

D. B's interval is half as wide. A and B each cover 95% in the simulations and cover about 95% of the time in the long run under the model.

Task 1 solution. Estimate mean produce spending

Quantity	Calculation and result	Meaning in context
Population parameter	μ, the mean spending on fresh produce among all households in the district during the specified week, in thousand COP.	The fixed, unknown district mean both teams want to estimate.
Point estimate	Both samples give $\bar{x} = 205$ thousand COP.	The same single estimate of the mean weekly produce spending from both samples.
Standard error	Using $\text{SE} = \sigma/\sqrt{n}$, Team A gets $50/\sqrt{25} = 10$ and Team B gets $50/\sqrt{100} = 5$, both in thousand COP.	Team B's sample mean varies less across repeated samples.
95% interval (Team A)	$\text{MoE} = z^* \text{SE} = 1.960(10) = 19.6$ thousand COP. The endpoints from 205 ± 19.6 give $[185.4, 224.6]$ thousand COP.	The wider range of plausible values for the district mean under the model.
95% interval (Team B)	$\text{MoE} = z^* \text{SE} = 1.960(5) = 9.8$ thousand COP. The endpoints from 205 ± 9.8 give $[195.2, 214.8]$ thousand COP.	The narrower range of plausible values for the same district mean under the model.



Task 2 solution. Separate precision from coverage

Correct option B. The widths are $2(19.6) = 39.2$ and $2(9.8) = 19.6$ thousand COP. Team B's fourfold sample size halves its standard error and therefore halves its MoE and interval width.

The observed rates are $192/200 = 96\%$ for A and $187/200 = 93.5\%$ for B. These rates can fluctuate across runs. Both procedures use the nominal level of 95% and cover when the distance from $\bar{x}$ to μ is at most $\text{MoE} = 1.960\,\text{SE}$. Under the model assumptions, that rule gives both procedures about 95% coverage in the long run. Each calculated interval either contains the mean or misses it.

